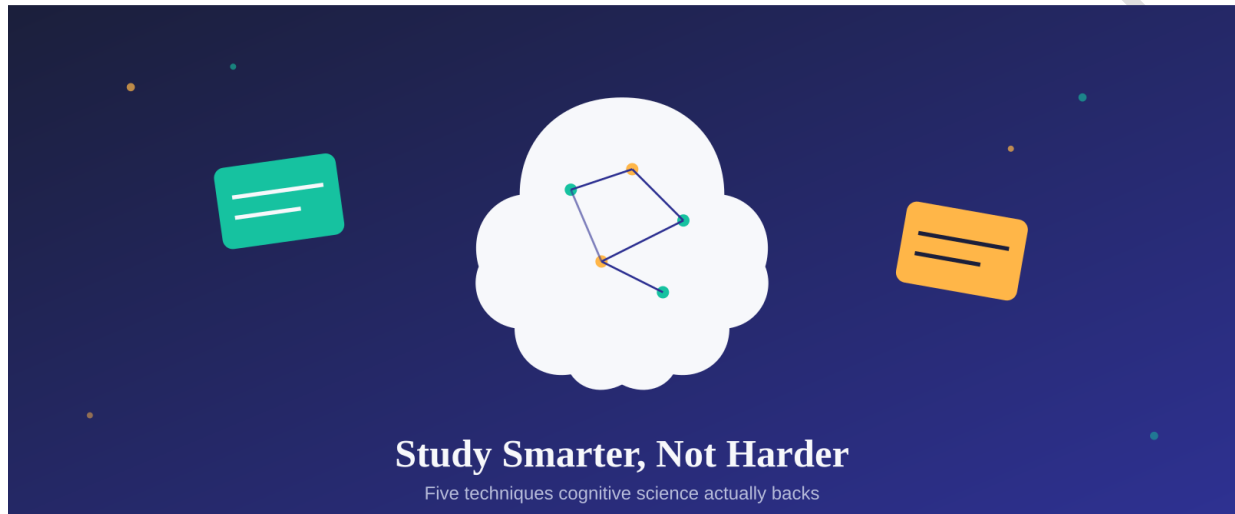


5 Study Techniques Backed by Cognitive Science That Actually Work

Discover evidence-based learning strategies to improve retention, focus, and academic performance.



Most students study the same way: reread the chapter, highlight the important-looking sentences, maybe rewrite the notes a bit neater. It feels productive. Unfortunately, decades of cognitive science research point to an uncomfortable truth these are some of the least effective ways to learn something.

The good news is that researchers have also identified exactly what does work techniques that are simple, free, and backed by real experimental evidence, not just study-hack folklore. Here are five of the strongest, and how to use them.

These aren't obscure academic findings, either they're some of the most replicated results in learning science, drawn from decades of controlled experiments across subjects ranging from vocabulary and foreign languages to physics and medicine. What varies is not whether they work, but how consistently students use them.

At Maieutic Edutech, we build these principles directly into how our platform paces lessons and reviews so consider this the science behind what you'll experience when you learn with us.

The Study Habits That Feel Productive but Aren't

In a landmark 2013 review of learning techniques, cognitive scientists rated the most commonly used study methods highlighting, rereading, and cramming as having low utility, despite being the most popular strategies students report using. It's worth understanding why these fall short before looking at what works:

- **Highlighting and underlining** direct attention to isolated phrases without requiring you to process how ideas connect, and studies find it provides little to no benefit over simply reading.
- **Rereading** builds familiarity with the text's appearance, which feels like understanding but is a different, and much weaker, mental process than being able to recall or apply it.
- **Cramming** can work for a recognition-based test the next morning, but the information decays fast, often within days, because it was never given time to consolidate.

THE PATTERN

All three feel effective in the moment because they're easy and create a sense of fluency. The five techniques below feel harder — and that extra effort is exactly what makes them work.



TECHNIQUE 01

Retrieval Practice — Force Your Brain to Recall, Not Reread

Also known as **active recall**, this is simply the act of closing the book and trying to pull information out of memory — through a practice quiz, a blank sheet of paper, or a flashcard — instead of passively reading it again.

Why it works: Every time you successfully retrieve a memory, you strengthen the neural pathway to it. Rereading creates a comforting illusion of familiarity — the material looks recognizable, so it feels learned — but recognition and recall are very different mental operations. Only recall is what you'll need on an exam or in a real conversation.

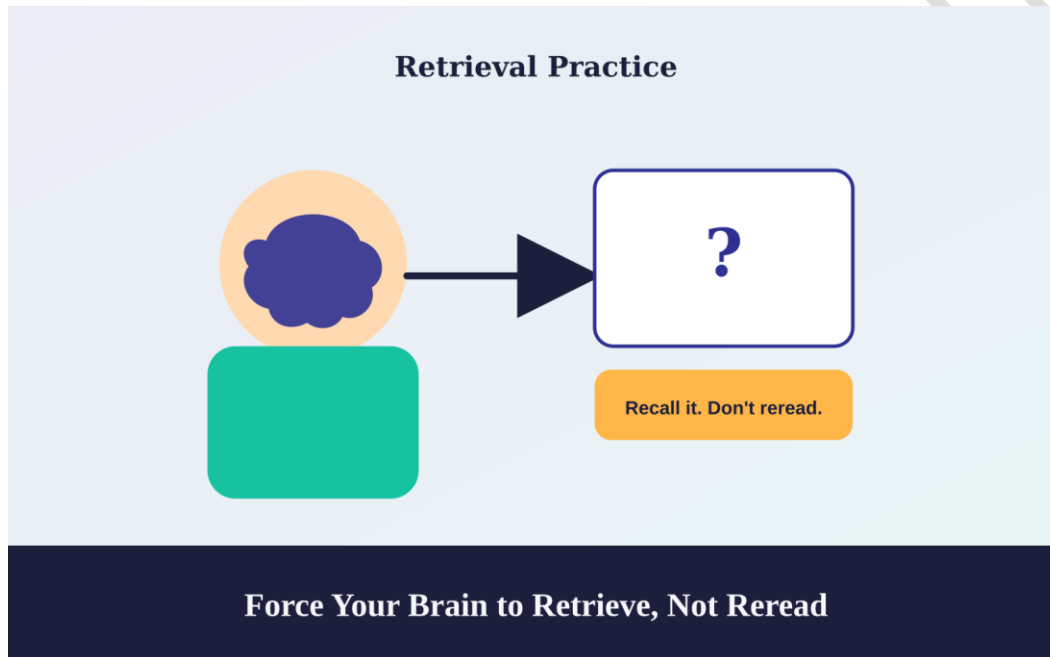
How to use it:

- Turn your notes into questions before you study, then close the notes and answer them from memory

- Use flashcards the right way — always try to answer before flipping the card, never flip first
- At the end of a reading session, put it away and write down everything you remember, unprompted

COMMON MISTAKE

Checking the answer the instant recall feels hard. Sit with the struggle for at least 10–15 seconds — that effortful searching is itself part of what strengthens the memory, even when you don't ultimately get the answer right.



TECHNIQUE 02

Spaced Repetition — Review Just Before You'd Forget

Cramming feels efficient because it produces short-term recognition fast. But memory naturally fades on a predictable curve — the *forgetting curve* — and cramming does nothing to interrupt it. **Spaced repetition** spreads review sessions out over increasing intervals, timed to catch the memory right as it starts to fade.

Why it works: Each time you successfully retrieve a memory right before you would have forgotten it, the memory gets consolidated more durably than it would from repeated same-day review. This is why studying for 20 minutes a day across two weeks beats one 5-hour cram session, even though the total time is similar.

How to use it:

- Review new material after 1 day, then 3 days, then a week, then roughly every two to four weeks after that
- Use a spaced-repetition flashcard tool that automatically schedules reviews based on how well you knew each card last time
- Don't wait until the night before an exam to start — spacing needs weeks, not hours, to work

COMMON MISTAKE

Treating every flashcard the same. Cards you already know cold should get pushed further out; cards you keep missing should come back sooner. Most spaced-repetition software does this adjustment for you automatically.



"Cramming produces recognition. Spacing produces memory that actually lasts."

TECHNIQUE 03

Interleaving — Mix Topics Instead of Blocking Them

Most students study in blocks: finish all the algebra problems, then move to geometry, then trigonometry.

Interleaving mixes related-but-different topics or problem types within a single study session instead.

Why it works: Blocked practice makes each problem easy to solve because you already know which method applies — you're on the algebra page, so you use algebra. Interleaving forces your brain to first identify which type of problem it's looking at, then choose the right method — which is exactly the skill you actually need on a real exam, where problems aren't sorted by chapter.

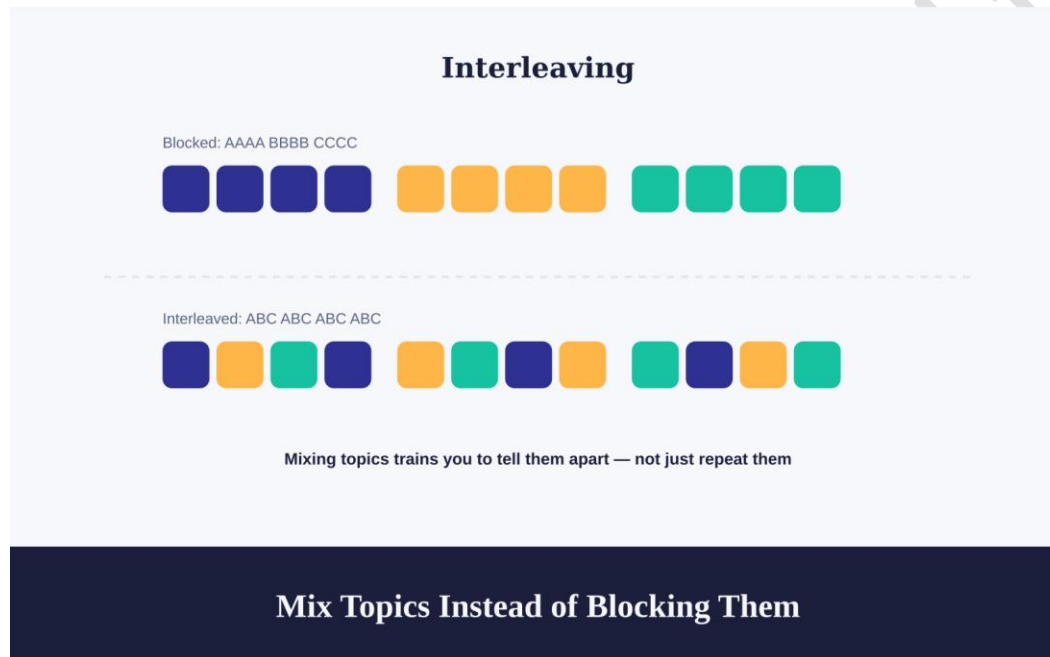
How to use it:

- When practicing problem sets, mix problem types together instead of doing 20 of the same kind in a row
- When studying multiple subjects for the week, rotate between them in a single sitting instead of dedicating one full day to each

- Expect it to feel harder and slower in the moment — that extra difficulty is exactly what produces better long-term learning

COMMON MISTAKE

Giving up on interleaving because a blocked practice session felt smoother. Smoother in practice often means weaker in memory — the fair comparison is how well you can tell the problem types apart a week later, not how it felt on the day.



TECHNIQUE 04

Elaboration & Self-Explanation — Explain It in Your Own Words

Elaboration means connecting new information to what you already know by continually asking why and how. Self-explanation is the practice of narrating your own reasoning out loud while you work through a problem or read a passage.

Why it works: New information sticks better when it's woven into a web of existing knowledge rather than stored as an isolated fact. Explaining a concept in your own words — or to an imaginary student who knows nothing about it — exposes the gaps in your understanding immediately, well before an exam does.

How to use it:

- After reading a concept, ask yourself “why is this true?” and “how does this connect to what I already know?”
- Try teaching the material out loud to an empty room, a study partner, or even a pet — if you get stuck, that's exactly where to focus your review

— For any formula or process, explain what would happen if one part of it changed



“If you can't explain it simply, you don't understand it yet — and now you know exactly where to focus.”

TECHNIQUE 05

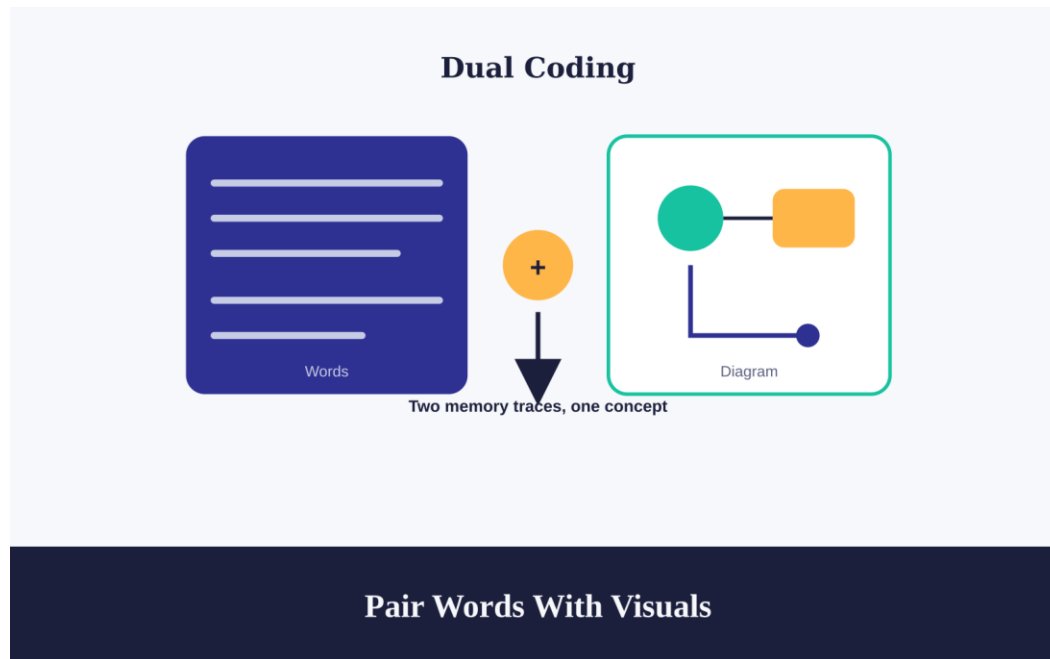
Dual Coding — Pair Words With Visuals

Dual coding means representing the same information in two different formats — words and a visual, like a diagram, timeline, or sketch — rather than just text alone.

Why it works: Verbal and visual information are processed through partly separate channels in memory. Encoding a concept both ways creates two retrieval paths back to the same idea instead of one, so if you forget the wording, the image can still cue the memory (and vice versa).

How to use it:

- When taking notes, sketch a simple diagram, timeline, or flowchart alongside the written explanation — it doesn't need to be a work of art
- Turn processes into flowcharts and relationships into simple maps instead of paragraphs whenever you can
- When reviewing, try reconstructing a diagram from memory, then check it against the original — this is retrieval practice and dual coding working together



Putting It All Together

None of these techniques require special tools, and they work even better combined. A realistic weekly rhythm might look like this:

1. **Turn new material into questions** the same day you first encounter it, instead of just rereading your notes.
2. **Schedule review sessions** at increasing intervals — a spaced-repetition app will do this automatically.
3. **Mix subjects and problem types** within a single sitting rather than studying one topic for hours straight.
4. **Explain each concept out loud** in your own words before you consider it “done.”
5. **Sketch it.** Pair at least one visual with any concept that has more than two moving parts.

It will feel harder than rereading and highlighting. That difficulty is the point — the techniques that feel the most effortful in the moment are, almost without exception, the ones that produce the strongest long-term learning.

Technique	In One Line
Retrieval Practice	Test yourself before rereading anything.
Spaced Repetition	Review at growing intervals, not all at once.
Interleaving	Mix topics and problem types together.
Elaboration	Explain the why and how in your own words.
Dual Coding	Pair every concept with a simple visual.

Where Maieutic Edutech Fits In

You shouldn't have to build a personal spaced-repetition schedule by hand or guess when to mix topics. That's exactly what **Maieutic Edutech's** adaptive learning engine handles for you — scheduling retrieval practice at the right intervals, interleaving related concepts automatically, and prompting you to explain ideas back before moving on.

Ready to study the way cognitive science recommends?

Study Techniques · Cognitive Science · Retrieval Practice · Spaced Repetition · Interleaving · Dual Coding

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